

QF632 Financial Data Science: Group Project

Objective

In the professional environment of quantitative research and financial data science, the most critical skill is often not executing a complex model, but rather identifying clean signals within unstructured or alternative datasets, formulating a testable economic hypothesis, and rigorously validating your statistical assumptions.

This project challenges you to act as an independent quantitative research team. You will identify an interesting, accessible data source, formulate your own financial or economic research questions, choose and justify your data-handling and modeling methodologies, and pitch your findings and operational constraints in a tight 15-minute presentation.

Core Project Milestones & Deliverables

To prevent "analysis paralysis" and ensure your data pipelines are viable, your team must clear three incremental milestones before the final presentation.

Milestone 1: The Feasibility Proposal (Due: May 30)

Submit a 1-page memorandum outlining your structural plan. Your proposal must explicitly answer:

- The Core Thesis: What is the fundamental financial, economic, or behavioral question you are investigating? What is the underlying market anomaly or corporate driver?
- Data Source & Access Protocol: Exactly what data sources will you use? How will they be accessed (e.g., API, web-scraping, Kaggle, or public files)?
- Proof of Ingestion: You must append a raw, printed snapshot of the first 5 rows of your ingested dataset to prove you have successfully bypassed access barriers or paywalls.

Milestone 2: Preprocessing & Exploratory Data Analysis (Due: June 6)

Before applying advanced predictive models, you must document your data engineering choices:

- Time-Series Structural Alignment: If you are combining alternative metrics with different tracking frequencies (e.g., daily web traffic with quarterly earnings or

weekly sentiment), explain how you match timestamps without introducing look-ahead bias or data leakage.

- **Scale Disparities & Missing Data:** Document how you handle missing values, anomalies, and scaling imbalances (e.g., standardizing unbounded financial variables versus tightly bounded percentage index arrays).

Milestone 3: Methodological Selection & Diagnostic Checks (Due: June 13)

Submit your preliminary modeling outputs alongside comprehensive statistical diagnostic proof. You must justify your math models based on the lessons covered in class:

- **If utilizing Linear Regression frameworks:** You must present a visual scatter plot of your residual tracking errors to test for homoscedasticity, address potential multicollinearity among correlated inputs, and flag any temporal autocorrelation patterns across consecutive periods.
- **If utilizing Principal Component Analysis (PCA):** You must present a diagnostic scree plot and visually justify where your "elbow" cutoff occurs. You must also interpret the loading vectors to explain the real-world economic themes captured by your top perpendicular components.

Final Deliverable: The 15-Minute “Quant Pitch”

The final deliverable of your research is a strict 15-minute presentation (followed by 5 minutes of Q&A). Your presentation should be structured as an institutional-grade investment pitch. It will take place on June 16.

Suggested Presentation Structure:

1. **The Mosaic Thesis:** State your research question clearly. Explain the economic logic or investor behavior (such as anchoring or leverage constraints) that makes your data source a predictive signal.
2. **The Data Pipeline & Enrichment:** Detail your data collection hurdles, time-series frequency matching, and how you combined traditional and alternative sets to build an informative data mosaic.
3. **Model Architecture & Diagnostic Proof:** Walk through your Linear Regression or PCA implementation. Do not just show performance metrics—you must show your diagnostic plots (scree plots, residual trends) to prove your statistical assumptions hold true.

4. Operational Realities: Address real-world execution challenges. How does factor cyclicity affect your strategy? How do trading turnover frequencies, bid-ask spreads, and transaction costs impact your theoretical gains? If your model targets small-cap assets, address execution liquidity and short-sale borrow constraints.
5. Q&A: Defend your data processing choices, structural assumptions, and conclusions.

Evaluation Criteria

Your project will be graded based on four equally weighted categories:

1. Creativity & Thesis Formulation: Originality of the data source selection and the economic rationale behind your research question.
2. Data Rigor & Preprocessing Integrity: Appropriate handling of time-series synchronization, scale normalization, and leakage prevention.
3. Methodological Competence & Diagnostics: Correct alignment of models to the problem, backed by transparent diagnostic validation (proving you checked your residuals, multicollinearity, or component thresholds).
4. Communication & Operational Awareness: Clarity and professionalism during the 15-minute presentation, with a realistic understanding of real-world trading friction, liquidity constraints, and execution costs.

Important Dates

- May 30: Milestone 1
- June 6: Milestone 2
- June 13: Milestone 3
- June 16: Presentation